

pancreatic-mets-kras-g12d-basal-arid1a-k9r3

Plain-language summary for patient and caregiver

Generated 2026-06-06

What this page is

Libby looked at your tumor's genetic profile and pulled together the treatment options that aim at the specific features driving this cancer. It then ranked them and wrote down both the promise and the problems with each one. This page is the plain-language version of that work. It is written for you and the people helping you decide, not for a specialist.

One thing to hold onto before you read further. This page only covers treatments that target the particular molecular features found in your tumor. The standard chemotherapy that almost everyone with this diagnosis is offered first is a separate conversation with your oncologist. We mention it below as the backbone that some of these trials build on, but we do not rank it here, because it is not a feature-targeted option and your care team owns that decision.

What we know about your cancer

You have pancreatic cancer (the ductal adenocarcinoma type) that has spread. There is a large mass in the pancreas itself, and several tumors larger than a centimeter in the liver. A gene-expression test put your tumor in the "basal-like" group, which tends to behave more aggressively than the other main subtype, and that matters when your team picks a chemotherapy partner.

The genetic testing turned up a few things that shape the options:

- A change in a gene called **KRAS**, the G12D version. This is the main engine of the cancer and the most useful target found. Most of the options below aim straight at it.
- A loss-of-function change in a gene called **ARID1A**. This is a second, separate target, less established but real.
- A single change in a gene called **MLH1**, which is one of the DNA-repair genes. On its own this can look like the kind of cancer that responds to immune-unleashing drugs, but the rest of your profile (the tumor is "microsatellite stable" and the overall mutation count is on the lower side) points the other way. So immune-checkpoint drugs on their own are probably low-yield here. There is a simple tissue test (called MMR protein staining, or IHC) that settles the question, and it is worth doing before anyone reaches for that kind of drug.
- A few other changes (in SMAD4, TP53, and some incidental genes) that mostly tell us about prognosis rather than pointing to a specific drug. One of them, in a gene called ASXL1, may actually be coming from your blood cells rather than the tumor; a paired blood test can sort that out so it is not mistaken for a tumor target.

Your tumor markers were read from a blood-based test (a liquid biopsy). Most of the trials will want the KRAS finding confirmed on an actual tissue sample before they enroll you, so getting a tissue biopsy looked at is a practical first errand.

What you told us matters most

You did not give us specific goals or priorities, so we set the balance between fighting hard and going easy on side effects right in the middle. You did tell us you are open to clinical trials, which matters a lot here, because every targeted option for this cancer is currently available only through a trial.

One honest flag: you are 79. When someone is in their late 70s, the side-effect burden of a treatment usually

weighs heavier in the decision than a neutral setting would capture. We have not tilted the ranking on a preference you did not state, but you will see this come up below, because the gentlest option and the best-studied option are not the same drug. If you and your oncologist decide that being easier on your body is the priority, the order of the top two could reasonably flip. That is a conversation worth having out loud.

The options

These are ranked, but the gaps between the top three are narrow and partly a matter of what you value. The fourth is a more distant possibility.

Option 1 — daraxonrasib (a KRAS-blocking pill)

This is the option to discuss first. It is a pill that blocks the KRAS protein driving your cancer. Of everything on this page, it has the strongest evidence behind it: a large randomized trial (about 500 people), where some got the drug and some got chemotherapy, so the comparison is real.

In that trial, run in people who had already had one prior treatment, the results were striking. People on daraxonrasib lived a median of about 13 months, compared with about 7 months on chemotherapy. Put another way, at the point the researchers analyzed the data, the drug roughly cut the risk of dying by about 60% relative to chemo. The cancer also shrank in about a third of people.

Here is the catch, and it is an important one. That trial was in people who had already had a first treatment. You would be starting this as your first treatment, and the trial testing it in that earlier setting (called RASolute 303) has not reported results yet. So the benefit at your stage is expected, based on how the drug worked later on, but not yet proven. The numbers above also come from a mix of KRAS subtypes; your exact G12D version was inside that group but was not analyzed on its own.

The main side effects are a skin rash and mouth sores, severe in roughly one in seven to one in eight people, plus the kind of general side effects any cancer drug brings (though fewer than chemo caused in the same trial). For an older person, the rash and mouth-sore problem matters, because dealing with them sometimes means cutting the dose, and a lower dose may mean less benefit. The trial team has a playbook for managing these, but it is the thing to watch.

This is a pill you take continuously. It fits your openness to trials, and the first-line trial accepts your KRAS type.

Option 2 — zoldonrasib (a more precisely targeted KRAS pill)

This option carries caveats. Here is the tradeoff to weigh. This is also a pill that blocks KRAS, but it is built to hit your exact G12D version specifically. It is the gentlest drug on this whole page. In an early study of about 40 people with your exact mutation, the cancer shrank in about 30%, and stayed controlled in about 80%. There were no severe (grade 4 or 5) side effects at the dose used; the side effects that did show up (nausea, some diarrhea, a little rash) were mostly mild.

So why is it second and not first? Because that 30% figure comes from an early study where everyone got the drug, with no comparison group, and only 40 people. We do not yet know how long the benefit lasts, because that part has not been measured and reported. A response rate without a sense of durability is a weaker kind of evidence than Option 1's randomized survival result. That is the open question here: real activity in your exact

mutation, but the staying power is unmeasured.

For a 79-year-old, the tolerability is a genuine point in its favor, and if going easy on your body is your priority, this is the one a careful person would put forward first. The first-line trial (RASolute 305) adds it on top of chemotherapy, and the combined side effects of that pairing have not been published on their own yet, so the marrow-suppression risk of the combination at your age is not yet a known quantity.

Option 3 — INCB161734 (another precise KRAS pill, given with chemo)

This option carries caveats. Here is the tradeoff to weigh. This is a third KRAS-blocking pill, made by a different company, also aimed at your G12D mutation. On the response numbers, it actually looks the strongest of the bunch: in an early study of about 41 people (all heavily pretreated), the cancer shrank in about 37%. As a single drug, its side effects stayed mild, and notably it did not cause the rash and mouth sores the others can.

The caveat is about how you would actually receive it. The first-line trial (DAWN-303) does not give this drug by itself. It stacks it on top of full chemotherapy. And in the early combination data, about 40% of people had a serious drop in their infection-fighting white blood cells. That is the concern for a 79-year-old: the chemo-plus-drug combination hits the bone marrow harder, and there is no dedicated safety study of that combination yet. The favorable side-effect profile is real for the drug alone, but the version you would enroll in bundles in the chemotherapy and its risks.

On the response numbers, the 37% here and the 30% for Option 2 are close enough that they are probably within the noise of two small studies, so this does not clearly beat Option 2 on effectiveness. It sits a notch lower mainly because the route you would take carries that documented marrow hit.

Option 4 — ceralasertib (a pill aimed at the ARID1A loss, a second target)

This option carries caveats. Here is the tradeoff to weigh. This one aims at the other target in your tumor, the ARID1A loss, through a completely different mechanism than the KRAS drugs. We keep it on the page because it is a genuinely independent second shot if the KRAS route does not work out. But it is the thinnest evidence here, and there are several real problems.

In the study that exists, across various ARID1A-deficient cancers, the cancer shrank in about 14% of people, and the responses were concentrated in gynecologic cancers. No pancreatic cancer responded in that data. So applying it to your pancreatic cancer is an educated guess based on the biology, not something that has been shown to work in this disease.

On top of that, getting in is not even certain. The trial requires a specific tissue stain (called BAF250a) to confirm the ARID1A protein is actually gone, and only about two-thirds of tumors with this gene change show that loss, so your genetic result alone does not guarantee eligibility. And the main trial for it is currently not enrolling. The drug's class is also known for lowering blood counts, especially anemia, which is harder to tolerate at 79.

This is here as a documented backup, not as something to act on now.

A note on immune therapies and KRAS vaccines

You may have read about immune treatments and KRAS vaccines for this kind of cancer, so it is worth saying plainly why they are not on the list. The published KRAS-targeted cell therapy that made headlines only works in

people with a particular immune-system type (a specific HLA marker), and your immune type is different, so it cannot recognize your tumor the way it needs to. The vaccine approaches that do not depend on immune type were studied in people whose cancer was caught early and largely removed by surgery, not in cancer that has spread the way yours has, so those results do not carry over to your situation. And the immune-checkpoint drugs are unlikely to help here for the DNA-repair reasons mentioned above. None of this is a target your KRAS pill options miss; it is just that those particular immune routes do not fit your case.

Questions to ask your oncologist

- For Option 1 (daraxonrasib), what is the plan for catching and managing the skin rash and mouth sores early, and at what point would we cut the dose?
- Since the first-line evidence for Option 1 has not reported yet, how much weight should we put on results from people who were further along in treatment than I am?
- For Options 2 and 3, the studies are small and early. What would we be watching for to know whether the drug is actually working for me, and how soon?
- Given that I am 79, should we lean toward the gentler drug (Option 2) even though it has less evidence behind it? How do you weigh that for me specifically?
- For Option 3, the trial gives the drug with chemotherapy, and about 40% of people had a serious drop in white blood cells. How would we monitor my blood counts, and what is the threshold for backing off?
- Can we get a tissue biopsy reviewed to confirm the KRAS finding, since most of these trials need that before I can enroll?
- Is it worth doing the MMR protein stain on my tumor to settle the immune-therapy question one way or the other?
- For the ARID1A option (Option 4), is the confirming stain (BAF250a) something we can order now so we know whether it is even a real fallback, and are any trials for it actually open?

Sources

The recommendations on this page draw on the following published studies and trial records.

Published studies (PubMed ID): 27958275, 35648703, 36216931, 37625401, 38593348, 40790272, 41667470, 41686845, 42223072.

Clinical trials (ClinicalTrials.gov ID): NCT03682289, NCT05726864, NCT06040541, NCT06179160, NCT06445062, NCT06625320, NCT07491445, NCT07522073, NCT07621718.